

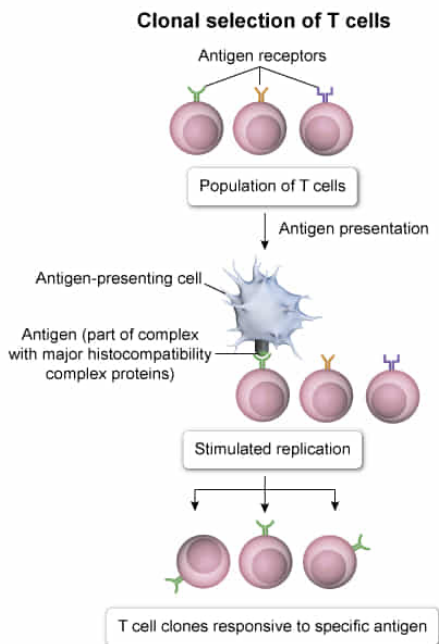
Clonal selection of B or T cells would require:

- ☐ A. a population of B cells, all of which express identical antigen receptors. [18%]
- ☐ B. a population of T cells producing a variety of antibodies, which are released into circulation. [11%]
- ☐ C. immune cells presenting antigens unable to bind a particular B cell's antigen receptor. [11%]
- ✓ ☒ D. immune cells presenting antigens specific for a particular T cell's antigen receptor. [58%]

Correct

58% answered correctly

Explanation:



The **immune system** employs various **types of defenses** to combat invading pathogens (eg, bacteria, viruses). **B lymphocytes** from the bone marrow and **T lymphocytes** from the thymus gland are mediators of humoral and cell-mediated immunity, respectively. These two types of immunity are the components of acquired (or adaptive) immunity, the body's highly tailored defense against specific pathogens.

Due to the wide variety of pathogen types, the potential number of different antigenic molecules is enormous, making the production of immune cells specific for the structure of each pathogen challenging. The adaptive immune system addresses this challenge through **clonal selection**, the **preferential replication** of B and T **lymphocyte clones that respond to a specific pathogen**.

The question asks for a requirement of clonal selection. **Clonal selection requires:**

- **A variety of B and T cell clones**, each consisting of one or a few cells with the same genotype but with slight variations in antigen receptors. This variation increases the likelihood that one or more of the clones will be able to respond to a specific pathogen.
- **Antigen receptors for a specific pathogen**. At least one of the B or T cells must possess a receptor that recognizes (ie, binds) an antigen from the pathogen.
- **Replication of antigen-responsive clones**. Antigen recognition stimulates replication of the responsive clone(s). Because these responsive clones are stimulated to replicate but clones unresponsive to the pathogen are not, the responsive clones represent a much greater proportion of the total B or T cell population.

(Choice A) Clonal selection in B cells involves the preferential replication, from a population of B cells with many different (*not* identical) antigen receptors, of those B cells whose antigen receptor matches (ie, binds) a specific antigen.

(Choice B) Plasma (*not* T) cells produce the antibodies that enter circulation.

(Choice C) Antigen presentation involves presentation of an antigen that is recognized (ie, bound) by an antigen receptor.

Educational objective:

Clonal selection is the process whereby B or T cell clones responsive to an antigen are stimulated to replicate and greatly increase in number in contrast to nonresponsive B or T cells, which do not increase in number.

Time spent:0 sec

Subject:
Biology

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System:
Skin and Immune Systems